

AKROMID® C3 GF 25 1 FR natural (7208)

Polyamide 66 + PA 6
AKRO-PLASTIC GmbH

Technical Data

Product Description

AKROMID® C3 GF 25 1 FR natural (7208) is an UL-listed, 25% glass fibre reinforced, heat stabilized and halogenfree flame retardant polyamide 6.6/6 - Blend listed at UL.

Components in mechanical engineering and the automotive industry where good flowability is required.

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E148915-10244324 • E532372-10217562		
Search for UL Yellow Card	• AKRO-PLASTIC GmbH • AKROMID®		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Filler / Reinforcement	• Glass Fiber, 25% Filler by Weight		
Additive	• Heat Stabilizer		
Features	• Flame Retardant	• Halogen Free	• Heat Stabilized
Uses	• Automotive Applications	• Engineering Parts	
Appearance	• Natural Color		
Resin ID	• PA66 + PA6 GF25 FR(40)		

Physical	Dry	Conditioned	Unit	Test Method
Density (23°C)	1.37	--	g/cm ³	ISO 1183
Spiral Flow				Internal Method
-- ³	22.0	--	cm	
-- ⁴	48.0	--	cm	
Molding Shrinkage				ISO 294-4
Across Flow	0.50 to 0.70	--	%	
Flow	0.10 to 0.30	--	%	

Mechanical	Dry	Conditioned	Unit	Test Method
Tensile Modulus	9200	6300	MPa	ISO 527-1/1
Tensile Stress (Break)	140	95.0	MPa	ISO 527-2/5
Tensile Strain (Break)	3.3	5.5	%	ISO 527-2/5
Flexural Modulus ⁵	8600	5900	MPa	ISO 178
Flexural Stress ⁵	225	160	MPa	ISO 178
Flexural Strain at Break ⁶	3.5	5.0	%	ISO 178

Impact	Dry	Conditioned	Unit	Test Method
Charpy Notched Impact Strength (23°C)	11	15	kJ/m ²	ISO 179/1eA
Charpy Unnotched Impact Strength				ISO 179/1eU
-30°C	74	--	kJ/m ²	
23°C	73	75	kJ/m ²	



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Thermal	Dry	Conditioned	Unit	Test Method
Deflection Temperature Under Load				
0.45 MPa, Unannealed	251	--	°C	ISO 75-2/B
1.8 MPa, Unannealed	225	--	°C	ISO 75-2/A
8.0 MPa, Unannealed	168	--	°C	ISO 75-2/C
Melting Temperature ⁷	262	--	°C	ISO 11357-3
CLTE				ISO 11359-2
Flow : 23 to 80°C	1.9E-5	--	cm/cm/°C	
Transverse : 23 to 80°C	1.0E-4	--	cm/cm/°C	
Temperature Index				IEC 60216
50% Loss of Tensile Strength, 20000 hr	130	--	°C	
50% Loss of Tensile Strength, 5000 hr	155	--	°C	
Electrical	Dry	Conditioned	Unit	Test Method
Electric Strength (3.00 mm)	12	--	kV/mm	IEC 60243-1
Comparative Tracking Index (Solution A)	> 600	--	V	IEC 60112
High Amp Arc Ignition (HAI)				UL 746A
0.40 mm	PLC 0	--		
0.8 mm	PLC 0	--		
1.6 mm	PLC 0	--		
3.2 mm	PLC 0	--		
Hot-wire Ignition (HWI)				UL 746A
0.40 mm	PLC 0	--		
0.8 mm	PLC 0	--		
1.6 mm	PLC 0	--		
3.2 mm	PLC 0	--		
Flammability	Dry	Conditioned	Unit	Test Method
Flame Rating				UL 94
0.8 mm	V-0	--		
1.6 mm	• V-0 • 5VA	--		
3.2 mm	• V-0 • 5VA	--		
Glow Wire Flammability Index				IEC 60695-2-12
0.40 mm	960	--	°C	
0.8 mm	960	--	°C	
1.6 mm	960	--	°C	
3.2 mm	960	--	°C	
Glow Wire Ignition Temperature				IEC 60695-2-13
0.40 mm	775	--	°C	
0.8 mm	750	--	°C	
1.6 mm	750	--	°C	
3.2 mm	775	--	°C	
Additional Information	Dry	Conditioned	Unit	Test Method
Reinforcement Content	25	--	%	ISO 1172



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Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ Mold Temperature: 90°C, Melt Temperature: 300°C, Injection Pressure: 750 bar, 1.00 mm

⁴ Mold Temperature: 90°C, Melt Temperature: 300°C, Injection Pressure: 750 bar, 2.00 mm

⁵ 2.0 mm/min

⁶ 2 mm/min

⁷ 10°C/min



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Where to Buy

Supplier

AKRO-PLASTIC GmbH

, Germany

Telephone: +49-2636-9742-0

Web: <http://www.akro-plastic.com/>

Distributor

Please contact the supplier to find a distributor for AKROMID® C3 GF 25 1 FR natural (7208)

